WORKED SOLUTION

STEP I 2002, Question 8

Pure • Loan Repayments and Geometric Series

STEP I 2002, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q8

The key idea

Write down what happens in a single year: the debt grows by a factor $1+\alpha$ and then the repayment is subtracted. Unrolling this one-step recurrence shows that each repayment R_k is "worth" $(1+\alpha)^{n-k}R_k$ by the end of year n, because it has stopped that much interest from building up. Everything else is a geometric series. For the final approximation, expand the powers of $1+\alpha$ by the binomial theorem, keeping just enough terms.

Solution

The general formula for C_{n+1}

At the end of year k the debt C_k becomes $(1+\alpha)C_k$ when interest is added, and then R_k is repaid. What is left is the debt at the start of year $k+1$:

$$C_{k+1} = (1+\alpha)C_k - R_k.$$

Applying this repeatedly,

$$\begin{aligned} C_{n+1} &= (1+\alpha)C_n - R_n \\ &= (1+\alpha)^2 C_{n-1} - (1+\alpha)R_{n-1} - R_n \\ &\vdots \\ &= (1+\alpha)^n C_1 - (1+\alpha)^{n-1}R_1 - (1+\alpha)^{n-2}R_2 - \cdots - R_n, \end{aligned}$$

so, since $C_1 = C$,

$$\boxed{C_{n+1} = (1+\alpha)^n C - \sum_{k=1}^{n}(1+\alpha)^{n-k}R_k}.$$

(To make this rigorous, check it holds for $n=1$, where it says $C_2 = (1+\alpha)C - R_1$, and that if it holds for n then multiplying by $1+\alpha$ and subtracting R_{n+1} gives it for $n+1$.)

Repayments $R_k = (1+\alpha)^k r$

Paying off the loan in N years means $C_{N+1} = 0$. With $R_k = (1+\alpha)^k r$, every term of the sum is the same:

$$(1+\alpha)^{N-k} R_k = (1+\alpha)^{N-k}(1+\alpha)^k r = (1+\alpha)^N r.$$

Hence

$$0 = C_{N+1} = (1+\alpha)^N C - N(1+\alpha)^N r = (1+\alpha)^N (C - Nr).$$

Since $(1+\alpha)^N \neq 0$, we get $C = Nr$, that is, $r = C/N$, as required.

This makes sense: the kth repayment is C/N "in today's money", grown by k years of interest, so altogether you repay the principal C plus exactly the interest on it.

Equal repayments R

Now $R_k = R$ for every k, and $C_{N+1} = 0$ gives

$$(1+\alpha)^N C = R \sum_{k=1}^{N} (1+\alpha)^{N-k} = R(1 + (1+\alpha) + \cdots + (1+\alpha)^{N-1}).$$

The bracket is a geometric series with first term 1, ratio $1+\alpha$ and N terms, so (as $\alpha > 0$, the ratio is not 1) it equals

$$\frac{(1+\alpha)^N - 1}{(1+\alpha) - 1} = \frac{(1+\alpha)^N - 1}{\alpha}.$$

Therefore $(1+\alpha)^N C = R\,\dfrac{(1+\alpha)^N - 1}{\alpha}$, and rearranging,

$$\frac{R}{C} = \frac{\alpha(1+\alpha)^N}{(1+\alpha)^N - 1},$$

which is the required result.

The case $\alpha = \frac{1}{50}$, $N = 4$

> **Ask yourself** How many terms of the binomial expansion should we keep, and why does the numerator need fewer than the denominator?

The denominator $(1+\alpha)^4 - 1$ starts at order α, while the numerator $\alpha(1+\alpha)^4$ is α times something starting at 1. So after cancelling a factor of α the numerator is $(1+\alpha)^4$ and the denominator is $((1+\alpha)^4 - 1)/\alpha$. To get both correct to first order in α we need $(1+\alpha)^4 \approx 1 + 4\alpha$ on top, but $(1+\alpha)^4 \approx 1 + 4\alpha + 6\alpha^2$ underneath (the $6\alpha^2$ becomes 6α after dividing by α). Thus

$$\frac{R}{C} = \frac{(1+\alpha)^4}{((1+\alpha)^4 - 1)/\alpha} = \frac{1 + 4\alpha + 6\alpha^2 + \ldots}{4 + 6\alpha + 4\alpha^2 + \alpha^3} \approx \frac{1 + 4\alpha}{4 + 6\alpha}.$$

With $\alpha = \frac{1}{50}$,

$$\frac{R}{C} \approx \frac{1 + \frac{4}{50}}{4 + \frac{6}{50}} = \frac{54}{206} = \frac{27}{103},$$

as required. The terms we dropped are of relative size about α^2, a fraction of a per cent here.

Checks

- With $N = 1$ the formula gives $R/C = 1 + \alpha$: borrow C, repay it with one year's interest. Correct.
- As $\alpha \to 0$, $(1+\alpha)^N - 1 \approx N\alpha$, so $R/C \to 1/N$: with no interest you simply repay C/N each year. This also agrees with $r = C/N$.
- The exact value is $R/C = \dfrac{6765201}{25760050} \approx 0.26262$, while $27/103 \approx 0.26214$; they differ by about 0.2%. Stepping the debt forward year by year with $R = 0.26262\,C$ leaves (to rounding) nothing owing after four years.

Common slips

- Getting the exponent wrong in the sum: R_k is paid at the end of year k, so it misses $n - k$ more years of interest, not $n - k + 1$.
- Approximating $(1+\alpha)^4 \approx 1 + 4\alpha$ in the denominator too: this gives $R/C \approx 27/100$, a cruder answer that doesn't match.

Beyond STEP

The quantity $\sum (1+\alpha)^{-k} R_k$ is the *present value* of the repayments, and the loan is cleared exactly when the present value of the repayments equals the amount borrowed; the formula for R is the standard annuity (mortgage) formula. Recurrences like $C_{k+1} = (1+\alpha)C_k - R_k$ are first-order linear difference equations, discrete cousins of the differential equation $y' = \alpha y - R(t)$, solved by the same "integrating factor" idea. For more problems that mix algebra with modelling in this way, see Siklos, *Advanced Problems in Mathematics*.

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